

Multiple Choice

1. **Answer:** A

Options: A) $[0.333, 0.2]$; B) $[0.1, 0.05]$; C) $[0.4, 0.3]$; D) $[0.5, 0.75]$; E) $[0.4, 0.5]$

Note: Correct option: A - $[0.333, 0.2]$

2. **Answer:** C

Options: A) The ring of complex numbers with rational coefficients; B) The ring of integers modulo 11; C) The ring of continuous real-valued functions on $[0, 1]$; D) The ring of rational numbers; E) The ring of complex numbers

Note: Correct option: C - The ring of continuous real-valued functions on $[0, 1]$

3. **Answer:** A

Options: A) 650; B) 700; C) 680; D) 655; E) 620

Note: Correct option: A - 650

4. **Answer:** A

Options: A) 17 over 2; B) 34 over 4; C) 16 over 2; D) 19 over 2; E) 33 over 4

Note: Correct option: A - 17 over 2

5. **Answer:** A

Options: A) -2; B) -5; C) -4; D) 0; E) 2

Note: Correct option: A - -2

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'is the mode of the data set'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 8. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 1.000. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key